

Report
Of
AWS Cloud Foundations: A Virtual Internship on
Cloud Computing Fundamentals

Submitted
To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: 2410031255
Name of Student: AMAN RAJ
Program: B.Tech CSE, Section 2CSE24, 4th Semester
Batch:
2026-27

CANDIDATE'S DECLARATION

I, AMAN RAJ, do hereby declare that the internship report titled "AWS Cloud Foundations: A Virtual Internship on Cloud Computing Fundamentals" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: AMAN RAJ

Roll No.: 2410031255

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the School of Computer Science and Engineering, IILM University, Greater Noida, for providing me the opportunity to undertake this virtual internship through the National Internship Portal, AICTE, Ministry of Education, in collaboration with EduSkills and AWS Academy. I am grateful to the AWS Academy team and my faculty mentors for their continuous guidance throughout the duration of this internship.

I express my gratitude to my parents for their blessings.

Signature

Student Name: AMAN RAJ

Roll No.: 2410031255

INTERNSHIP COMPLETION CERTIFICATE



Chapter No.	Chapter Name	Page Nos.
	Cover Page	----
1.	Candidate's Declaration	i
2.	Acknowledgment	ii
3.	Internship Completion Certificate	iii
4.	Project Description 4.1 Introduction 4.2 Organization Profile 4.3 Problem Statement 4.4 Project Objectives 4.5 Scope of the Project 4.6 Technologies and Tools Used 4.7 System Architecture 4.8 Methodology 4.9 Expected Outcomes 4.10 Certificates of completion and other communication mail proof	1
5.	Bibliography/References	

4. Project Description

4.1 Introduction

Cloud computing has emerged as one of the most transformative technologies in the modern IT industry, enabling organizations to access computing resources such as servers, storage, databases, networking, and software over the internet without owning physical infrastructure. As part of the B.Tech CSE curriculum at IILM University, Greater Noida, I undertook a 10-week virtual internship on 'Cloud Foundations', offered through the National Internship Portal of AICTE, Ministry of Education, in collaboration with EduSkills and AWS Academy, during the period April-June 2026.

The internship curriculum was designed and delivered using AWS Academy learning resources and provided a structured, hands-on introduction to core cloud computing concepts and the Amazon Web Services (AWS) ecosystem. The program covered foundational topics including cloud computing models, AWS global infrastructure, Identity and Access Management (IAM), compute services (Amazon EC2), storage services (Amazon S3 and EBS), networking (Amazon VPC), databases, monitoring, security, and cost management. This report documents the learning outcomes, concepts studied, and practical exposure gained during the course of the internship.

4.2 Organization Profile

The internship was facilitated by EduSkills Foundation, a not-for-profit organization working under the 'Nation Building Through Skills' initiative, in partnership with AICTE (All India Council for Technical Education) under the Ministry of Education, Government of India, through the National Internship Portal. The technical curriculum for this internship was provided by AWS Academy, an initiative of Amazon Web Services (AWS) that equips students with industry-relevant cloud computing skills through structured coursework, hands-on labs, and assessments.

AWS (Amazon Web Services) is the world's leading cloud service provider, offering over 200 fully featured services from data centers globally, used by millions of customers including start-ups, large enterprises, and government agencies to lower costs, become more agile, and innovate faster. AWS Academy's curriculum is designed

by AWS subject matter experts and aligned with industry certification pathways such as the AWS Certified Cloud Practitioner.

4.3 Problem Statement

Traditional on-premises IT infrastructure requires significant upfront capital investment, dedicated physical space, manual scaling, and continuous maintenance, which limits the agility and scalability of organizations, particularly students and early-stage developers who wish to build, test, and deploy applications without owning hardware. There existed a clear gap in practical, hands-on exposure to industry-standard cloud platforms within the traditional academic curriculum. This internship was undertaken to address that gap by providing structured, guided exposure to real-world cloud computing concepts and the AWS platform, enabling learners to understand how modern applications are designed, deployed, and managed on the cloud.

4.4 Project Objectives

- To understand the fundamental concepts of cloud computing, including its deployment models (public, private, hybrid) and service models (IaaS, PaaS, SaaS).
- To gain hands-on familiarity with AWS Management Console and core AWS services.
- To learn the principles of Identity and Access Management (IAM) for securing cloud resources.
- To understand AWS compute services, primarily Amazon EC2, and virtual server provisioning.
- To explore AWS storage solutions such as Amazon S3 and Amazon EBS.
- To learn the basics of networking on AWS through Amazon Virtual Private Cloud (VPC).
- To understand AWS's approach to security, compliance, monitoring, and cost management.
- To build a foundation for pursuing the AWS Certified Cloud Practitioner certification.

4.5 Scope of the Project

The scope of this internship was limited to foundational, conceptual, and lab-based learning of cloud computing using the AWS platform, as structured under the AWS Academy Cloud Foundations curriculum. It did not involve the deployment of a production application but focused on building a strong theoretical and practical base across core AWS service categories. The learning was carried out through self-paced modules, instructor-led virtual sessions, hands-on labs conducted in the AWS Academy Learner Lab sandbox environment, and periodic knowledge assessments.

4.6 Technologies and Tools Used

- Amazon Web Services (AWS) Management Console
- Amazon EC2 (Elastic Compute Cloud) — virtual servers
- Amazon S3 (Simple Storage Service) — object storage
- Amazon EBS (Elastic Block Store) — block storage volumes
- AWS IAM (Identity and Access Management)
- Amazon VPC (Virtual Private Cloud) — networking and subnets
- Amazon RDS — managed relational databases (conceptual overview)
- AWS CloudWatch and AWS Billing & Cost Management Console
- AWS Academy Learner Lab (sandbox training environment)

4.7 System Architecture

The AWS Cloud Foundations curriculum introduces learners to the layered architecture of AWS's global infrastructure, which consists of Regions, Availability Zones, and Edge Locations, ensuring high availability, fault tolerance, and low latency for hosted applications. A typical foundational AWS architecture, as studied during the internship, consists of the following components:

- A Virtual Private Cloud (VPC) providing an isolated virtual network with public and private subnets.
- Amazon EC2 instances deployed within subnets to host compute workloads, secured using Security Groups acting as virtual firewalls.

- Amazon S3 buckets used for durable, scalable object storage of static assets and backups.
- IAM roles, users, and policies governing fine-grained access control to AWS resources.
- CloudWatch used for monitoring resource utilization, logs, and setting alarms.

This layered architecture illustrates the shared responsibility model of cloud computing, wherein AWS manages the security 'of' the cloud (physical infrastructure), while the customer is responsible for security 'in' the cloud (data, access management, and configuration).

4.8 Methodology

The internship followed a structured, module-based methodology spread across 10 weeks, combining self-paced e-learning, instructor-led sessions, and hands-on labs:

- Week 1-2: Introduction to Cloud Computing — concepts, benefits, deployment and service models.
- Week 3-4: AWS Global Infrastructure and Identity and Access Management (IAM).
- Week 5-6: Compute services — Amazon EC2, instance types, and Auto Scaling concepts.
- Week 7: Storage services — Amazon S3, EBS, and data lifecycle management.
- Week 8: Networking — Amazon VPC, subnets, route tables, and security groups.
- Week 9: Databases, monitoring (CloudWatch), and security & compliance fundamentals.
- Week 10: Pricing and cost management, final assessment, and certification preparation.

Each module was reinforced through hands-on labs conducted in the AWS Academy Learner Lab, a safe sandbox environment that allows learners to practically apply concepts without incurring real costs or affecting production systems. Knowledge checks and quizzes were completed at the end of each module to reinforce learning.

4.9 Expected Outcomes

- A strong conceptual understanding of cloud computing fundamentals and the AWS ecosystem.
- Practical familiarity with provisioning and managing core AWS services such as EC2, S3, IAM, and VPC.
- Ability to design a basic, secure cloud architecture following the AWS Well-Architected Framework principles.
- Readiness to pursue the AWS Certified Cloud Practitioner certification.
- Enhanced employability through exposure to an industry-recognized cloud platform, complementing academic learning in Computer Science and Engineering.

4.10 Certificates of Completion and Other Communication/Mail Proof

The Certificate of Virtual Internship, issued jointly by AICTE (Ministry of Education), the National Internship Portal, and EduSkills, with curriculum provided by AWS Academy, is attached in Chapter 3 of this report (Internship Completion Certificate). The certificate confirms successful completion of the 10-week Cloud Virtual Internship conducted during April-June 2026, bearing Certificate ID: 4ddd8c62371685bd2485 and Student ID: STU6998870df2e311771603725.

5. Bibliography/References

1. Amazon Web Services. AWS Academy Cloud Foundations Curriculum. AWS Academy, 2026.
2. Amazon Web Services. AWS Documentation. <https://docs.aws.amazon.com/>
3. EduSkills Foundation. National Internship Portal, AICTE, Ministry of Education, Government of India. <https://internship.aicte-india.org/>
4. Amazon Web Services. AWS Well-Architected Framework. AWS Whitepapers, 2026.